# Regular Elements as Twisted Principal Series

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## 1 Centralizers

Let $G$ and $H$ be reductive groups over a local field.

In Langlands, *functoriality* is the principle that a map ${}^L H \to {}^L G$ induces a *transfer*

$$\left\{ \begin{array}{c} \text{admissible irreps} \\ \text{of } H \end{array} \right\} \to \left\{ \begin{array}{c} \text{admissible irreps} \\ \text{of } G \end{array} \right\}.$$

One of the basic testing grounds was and is *endoscopy*.

Roughly, $H$ is *endoscopic* to $G$ iff ${}^L H = C_{{}^L G}(s)^\circ$ for some semisimple element $s \in {}^L G$.

Today's story is **not** an analogue of endoscopy for Weyl groups, but...

Let $W$ be a well-generated (complex) reflection group.

(Springer '74)

An element $w \in W$ is *regular* iff it has a regular eigenvector in the reflection rep $V$.

In this case, $C_W(w)$ is also a reflection group.

Example The identity $e$.

Example For $W$ real, the longest element $w_\circ$.

Example For $W$ irreducible, Coxeter elements.

(Broué–Michel '97)

$w$ is regular iff its minimal positive braid lift has a central power.

Now take $W$ to be the Weyl group of a finite reductive group

$$G = \mathbf{G}^F, \quad \text{where } F \text{ is a } q\text{-Frobenius map.}$$

A choice of $F$-stable, maximally $F$-split $\mathbf{A} \subseteq \mathbf{G}$ yields:

$$\begin{array}{rcl} \{\text{maximal tori of } G\}/\text{conj.} & \simeq & W/\text{conj.} \\ T & = & \mathbf{A}^{wF} \end{array}$$

(Deligne–Lusztig '76)

All irreps of $G$ arise from such $T$ via *cohomological* induction maps

$$R_T^G \colon K_0(T) \to K_0(G).$$

Special features when $T$ is of type $[w]$ for *regular* $w$.

An irrep of $G$ is *unipotent* iff it belongs to

$$\mathrm{Uch}_T^G = \{\rho \mid (\rho, R_T^G(1))_G \neq 0\} \quad \text{for some } T.$$

For such $\rho$, a *generic degree* $\mathrm{Deg}_\rho(x) \in \mathbf{Q}[x]$ such that

$$\boxed{\deg \rho = \mathrm{Deg}_\rho(q).}$$

(Lusztig, Broué–Malle–Michel '93)

If $w$ is regular of order $m$, and $T$ is of type $[w]$, then

$$\mathrm{Uch}_T^G = \{\text{unipotent } \rho \mid \mathrm{Deg}_\rho(\zeta_m) \neq 0\}.$$

And $\exists\, \chi_T^G \colon \mathrm{Uch}_T^G \xrightarrow{\sim} \mathrm{Irr}(C_W(w))$ such that

$$\boxed{\deg \chi(\rho) = |\mathrm{Deg}_\rho(\zeta_m)|.}$$

Proved case by case. Morally, why the bijection?

Conjecture (Broué–Malle '93) $H_T^G(x)$ such that

$$\boxed{\begin{aligned} H_T^G(q) &\simeq \mathrm{End}_G(\mathrm{H}^*_{\mathrm{c,\acute{e}t}}(X_w)),\\ H_T^G(\zeta_m) &\simeq \mathbf{C}C_W(w). \end{aligned}}$$

$\chi_T^G$ should arise from a double centralizer theorem.

Example Write $A := \mathbf{A}^F$. It is of type $[e]$.

$$X_e = (\mathbf{G}/\mathbf{B})^F = G/B.$$

$R_A^G(1)$ consists of the *unipotent principal series*: the irreps occuring in $\mathbf{C}[G/B]$.

$H_A^G(x)$ is the usual Hecke algebra of $W$.

**Example** Take $G = \mathrm{Sp}_4(\mathbf{F}_q)$ and $T$ of *Coxeter type*.

Any Coxeter element $c$ has order $h = 4$.

| | $\mathrm{Deg}_\rho(x)$ | $\mathrm{Deg}_\rho(1)$ | $\mathrm{Deg}_\rho(\zeta_4)$ |
|---|---|---|---|
| | $1$ | $1$ | $1$ |
| | $\frac{1}{2}x(x+1)$ | $1$ | $\cdot$ |
| | $\frac{1}{2}x(x+1)^2$ | $2$ | $-1$ |
| | $\frac{1}{2}x(x+1)$ | $1$ | $\cdot$ |
| | $x^4$ | $1$ | $1$ |
| $\theta_{10}$ $\rightarrow$ | $\frac{1}{2}x(x-1)^2$ | $\cdot$ | $1$ |

$W = C_W(e)$ is dihedral of order 8, while $C_W(c)$ is cyclic of order 4.

Anything special about $\mathrm{Uch}_A^G \cap \mathrm{Uch}_T^G$?

**($\approx$ Lusztig '78)** For $G$ almost simple and $\rho \in \mathrm{Uch}_A^G$,

$$\mathrm{Deg}_\rho(\zeta_h) = \begin{cases} (-1)^k & \chi_A^G(\rho) = \Lambda^k(V), \\ 0 & \text{else.} \end{cases}$$

So for $T$ of Coxeter type, $\chi_A^G$ restricts to

$$\mathrm{Uch}_A^G \cap \mathrm{Uch}_T^G \xrightarrow{\sim} \{\Lambda^k(V) \mid 0 \le k \le r\},$$

where $r = \mathrm{rank}(G)$.

For $W = S_n$, indirectly recovers Frobenius's result:

$$\chi(n\text{-cycle}) = \begin{cases} (-1)^k & \chi = \Lambda^k(V), \\ 0 & \text{else.} \end{cases}$$

## 2 Combinatorics

$\mathrm{Uch}_A^G \cap \mathrm{Uch}_T^G$ is implicit throughout my work with Nathan.

For any word $\vec{s}$ in simple reflections, write

$$\vec{w} \nearrow \vec{s} \quad \text{iff} \quad \vec{w} \text{ is a Deodhar walk for } \vec{s}.$$

This means a walk in $W$, starting and ending at $e$, such that for all $i$:

- either $w_{i+1} = w_i s_{i+1}$ or $w_{i+1} = w_i$.
- $w_{i+1} \leq w_i s_{i+1}$. *If you can descend, you must.*

Let $d_{\vec{w}}$, *resp.* $e_{\vec{w}}$, count the descents, *resp.* stays.

These appear in point counts of Deodhar cells.

(Galashin–Lam–T.–Williams '22)

For any *Coxeter word* $\vec{c}$, and $p > 0$ coprime to $h$,

$$\frac{1}{q^{rp/2}(1-q)^r} \sum_{\vec{w} \nearrow \vec{c}^p} q^{d_{\vec{w}}} (q-1)^{e_{\vec{w}}}$$

$$= \sum_{\chi \in \mathrm{Irr}(W)} \mathrm{Deg}_\chi(\zeta_h)(1, [\Delta_{p/h}(\chi)]_q)_W$$

$$= (1, [L_{p/h}(1)]_q)_W \quad \leftarrow \; q\text{-Catalan}$$

where:

- $\mathrm{Deg}_\chi = \mathrm{Deg}_\rho$ when $\chi_A^G(\rho) = \chi$.
- $\Delta$, *resp.* $L$, denotes Verma, *resp.* simple, rational Cherednik modules.
- $[M]_q = \sum_i \mathrm{gr}_i \, q^{i/2} M_i \in K_0(W)((q))$.

In general:

$\Delta$'s have easy characters, but are infinite-dimensional.

$L$'s are finite-dimensional, but their characters are mysterious.

(Berest–Etingof–Ginzburg, Chmutova–Etingof '03)

$$\sum_k (-1)^k [\Delta_{p/h}(\Lambda^k(V))]_q = [L_{p/h}(1)]_q.$$

Recall that $\mathrm{Deg}_\chi(\zeta_h) = \begin{cases} (-1)^k & \chi_A^G(\rho) = \Lambda^k(V), \\ 0 & \text{else.} \end{cases}$

If $p/h$ simplifies to $d/m$ in lowest terms:

How does $\sum_\chi \mathrm{Deg}_\chi(\zeta_m)[\Delta_{d/m}(\chi)]_q$ expand into $L$'s?

## 3 Cohomology

Let $\hat{G}$ be the associated form over $\mathbf{C}(\!(z)\!)$. Again,

$$\{\text{maximal tori of } \hat{G}\}/\text{conj.} \quad \simeq \quad W/\text{conj.}$$

$\gamma \in \mathrm{Lie}\,\hat{G}$ is *homogeneous* of slope $d/m$ iff

$$\gamma(c^m z) \underset{\text{conj.}}{\sim} c^d \gamma(z) \quad \text{for } c \in \mathbf{C}^\times.$$

For such $\gamma$ generic* enough, $C_{\hat{G}}(\gamma)$ is a maximal torus whose type is regular of order $m$.

Example Take $\hat{G} = \mathrm{SL}_2(\mathbf{C}(\!(z)\!))$:

$\begin{pmatrix} z^d & \\ & -z^d \end{pmatrix}$ has slope $d$; $\begin{pmatrix} & 1 \\ z^d & \end{pmatrix}$ has slope $d/2$.

* regular semisimple

For generic homogeneous $\gamma$ of slope $d/m$, the

$$\textit{affine Springer fiber} \quad \mathcal{F}l_\gamma \subseteq \mathcal{F}l_{\hat{G}}$$

has a $\mathbf{C}^\times$-action studied by Lusztig–Smelt, Fan, Sage, Sommers, Varagnolo–Vasserot, ...

(Oblomkov–Yun '16) Cherednik action* on

$$E_\gamma = \mathrm{gr}_*^{\mathsf{P}} \mathrm{H}^*_{\mathbf{C}^\times,\mathrm{loc}}(\mathcal{F}l_\gamma)_{\mathrm{st}}.$$

Let $T \subseteq G$ be regular of order $m$, like $C_{\hat{G}}(\gamma)$.

Conjecture (T.–Xue '23) $\mathrm{End}_{\mathrm{RCA}}(E_\gamma) \simeq H_T^G(1)$,

$$[E_\gamma]_q = \sum_{\rho \in \mathrm{Uch}_A^G \cap \mathrm{Uch}_T^G} [\Delta_{d/m}(\chi_A^G(\rho))]_q \otimes \chi_T^G(\rho)|_{x\to 1}.$$

* assuming $\gamma$ elliptic

Theorem (T.–Xue '23) The conjecture holds when:

- $m$ is the (twisted) Coxeter number of $\hat{G}$.
- $(\hat{G}, m) = ({}^2A_2, 2), (C_2, 2), (G_2, 3), (G_2, 2)$.

The mysterious analogy:

| | | |
|---|---|---|
| $\mathbf{F}_q$ | $(q, q)$ | $\mathrm{H}^*_c(X_e) \otimes_G \mathrm{H}^*_c(X_w)$ |
| $\mathbf{C}(\!(z)\!)$ | $(\zeta_m, 1)$ | $E_\gamma$ |

Out of time to explain how $E_\gamma$ is *level-rank duality* between $H_A^G(\zeta_m)$ and $H_T^G(1)$.

*Thank you for listening.*